NANYANG TECHNOLGICAL UNIVERSITY
School of Electrical and Electronic Engineering
IE4728 Enterprise Network Design

Project description of an Enterprise Network

1. Design requirements:
    a. The School of EEE has recently added two buildings, S2.1 and S2.2 to expand its research and lab facilities. The facilities are to be supported with an enterprise network to facilitate interworking, file and data sharing, printing.
    b. The enterprise network interconnects local area networks (LANs) in the various labs and staff/admin offices through a wide area network (WAN).
    c. The network supports interworking among four workgroups, namely Academic staff (Acad), Administrative staff (Admin), Research Students (RS), and Lab PC users (Lab).
    d. Except for the Lab workgroup, users belonging to the same workgroup collaborate with other members across different network segments.
    e. User groups belonging to the Lab workgroup do not interoperate nor share resources with other labs.
    f. The number of users in the various labs and offices are indicated in Figure 1, which shows the cross-sectional view of the lab and office layouts in the two buildings.
    g. The number of shared printers and servers are not reflected in Figure 1 and must be worked out in the design and for all user groups.
    h. Each user group in a lab/office must have its own printer(s). More than one printer should be provided for users in a large physical workspace.
    i. Each workgroup, except the Lab workgroup must have also at least a server, which may be shared across other workgroups. Lab servers are not shared.
    j. Access Control List (ACL) is implemented to manage access to network resources and interworking of workgroups.
    k. All user PCs, servers and printers on the network must be assigned an IP address each.
    l. To conserve network address space, a network may be segmented into smaller LAN segments; whilst small segments may be augmented into a larger LAN segment to minimize address wastage.
    m. A LAN segment must not span beyond its maximum network diameter. For convenience, the distance spanning <u>two adjoining</u> labs/offices shall constitute the maximum diameter. Adjoining rooms are neighboring rooms sharing a common wall or between floors.
    n. Access to campus wide services and the Internet by various workgroups is an integral function of the enterprise network. This includes access to HTTP (Port 80), DNS (Port 53), and SQL (Port 1433) services.
    o. In addition, access to the following services and shared resources are allowed.
        i.) Acad and Admin share a SNMP server on Port 25 for email communication.
        ii.) RS and Acad share FTP service on Port 21 for file transfer.
        iii.) Every workgroup has a SMB server on Port 445 for file-sharing, except for the Lab workgroup. Note that a server may host multiple services.
    p. However, restricted access is imposed for the following scenarios:
        i.) Laboratory users (Lab) <u>only permitted</u> to access HTTP service on the enterprise network. This means access to any of the other lab/office networks are not allowed.
        ii.) RS workgroup is <u>not permitted</u> to access the email service shared between Acad and Admin, although RS can establish FTP service with the Acad workgroup.
        iii.) Admin workgroup is <u>not permitted</u> to access Acad and RS servers, although Acad workgroup is permitted to access the Admin server for file-sharing and the RS server for file transfer service.
        iv.) All external access into the campus networks is <u>not allowed</u>, except for HTTP service on the enterprise network.
    q. The enterprise network should be designed using VLSM (variable length subnet mask) subnetting to minimize IP address wastage and subnets should be assigned in such a way that maximizes route aggregation in supernets.
    r. Other network services, such as DHCP and RIP v2 may be considered to facilitates efficient implementation of the enterprise network.
    s. The network is to be designed and built based on TCP/IP architecture.

2. Design work:
   a. The network design involves a top-down analysis of the requirements as described above.
   b. The design of the network is implemented bottom-up using five table templates, Table 1 to 5.
   c. The implementation should minimize IP address wastage and maximize IP route aggregation.
   d. The design should have a high degree of modularity and exhibit a layered hierarchy.
   e. All layers of the network should be optimized for robustness against node and link failures.
   f. All routers on the distribution layer should have even load distribution. For convenience, router load shall consider only static loading factors, such as number of connected subnets, memory and processing requirements on each interface.
   g. Servers should be placed at appropriate network layers to prevent network flooding.
   h. You should use the table templates as worksheets to work out the design:
      i.) Table 1, work out the network addresses for all subnets and supernets in the network.
      ii.) Table 2, determine the host addresses for all PCs, router interfaces, printers and servers.
      iii.) Table 3, work out the IP addresses of all servers and their attached router interfaces.
      iv.) Table 4, identify the router and its interface for placing access control lists (ACLs).
      v.) Table 5, write the access-list statements for each ACL identified in Table 4.
   i. The network design should be implemented using the "Packet Tracer" simulator.
      i.) Label all hosts and devices so that they can be similarly identified in the design report.
      ii.) For each LAN segment, assign the first useable host address to the router interface followed by the first and last consecutive PC host addresses. The server should be assigned the last useable address and printer address in-between the remaining useable address range of the subnet.
      iii.) Write the access-control lists (ACLs) using the CLI in Packet Tracer and apply them to appropriate router interfaces. Maintain a record of the router interfaces and where ACLs have been configured.
   j. <u>Importance</u>: ensure subnetting plan is fully configured and functional before proceeding to implement access control lists (ACLs) on the network. This ensures any issues with the implementation of ACLs will not be complicated by underlaying issues with the subnetting plan.

3. Project Submission:

   Project submission includes design report, design simulation, and power-point slides should be submitted on the course site. Label the filename using the format, "Fxx-Gyy_Title"; where Fxx is your tutorial group number, such as F53, and Gyy is the group number of the project team, such as G01. Title corresponds to 'Report', 'Simulation', and 'Slides'.
   a. <u>Design Report</u>.

      A soft copy of the report in Microsoft Word is to be submitted to Turnitin on the course site.
      i.) The report should contain at least the following chapters:
         (1) Introduction and design objectives
         (2) Top-down analysis of network requirements
            - LAN segmentation of physical networks (T1)
         (3) Bottom-up network implementation of logical networks
            Host IP address assignment (T2)
         (4) Multilayered network design
            - Hierarchical layered design and servers placement (T3)
         (5) Network security requirements
            - Access control list filtering (T4)
         (6) Simulation and Testing
            - Scenarios for network interconnectivity and access control lists (T5)
         (7) Summary and conclusion
         (8) Members' contribution to the contents in the above chapters.
      ii.) All good designs should be appropriately justified and substantiated.
      iii.) Reports are assessed for quality of presentation and readability, justification of network design with respect to modularity and functional layers, correctness and completeness of IP address assignments, access control lists, and simulation scenarios.

   b. <u>Design Simulation</u>.

      A soft copy of Packet Tracer simulation of the network design to be submitted on the course site.
      i.) Provide working simulations of the designed network.
      ii.) Provide simulation scenarios to verify all design requirements as described in Section 1.
      iii.) Provide a brief description of the simulation process for each scenario.

c. Design Slides.

A soft copy of the design slides in PPT format must be submitted on the course site.

i.) Present an executive summary of the network design in the form of presentation slides, highlighting the design approach and key features.

ii.) Provide justifications supporting the effectiveness of a good network design.

iii.) Highlight the key selling points of the network design in the slides.

iv.) The slides will be graded as part of the report; no oral presentation is required.

4. Submission Deadline:

a. All submissions are due one week after the last day of the 3-week project design class.

b. Penalty will be imposed for late submissions. Penalty marks of 30%, 50%, and 70% will be imposed on late submissions by 1 day, 2 days, and 3 days, respectively. No mark will be given for reports submitted later than 3 days.

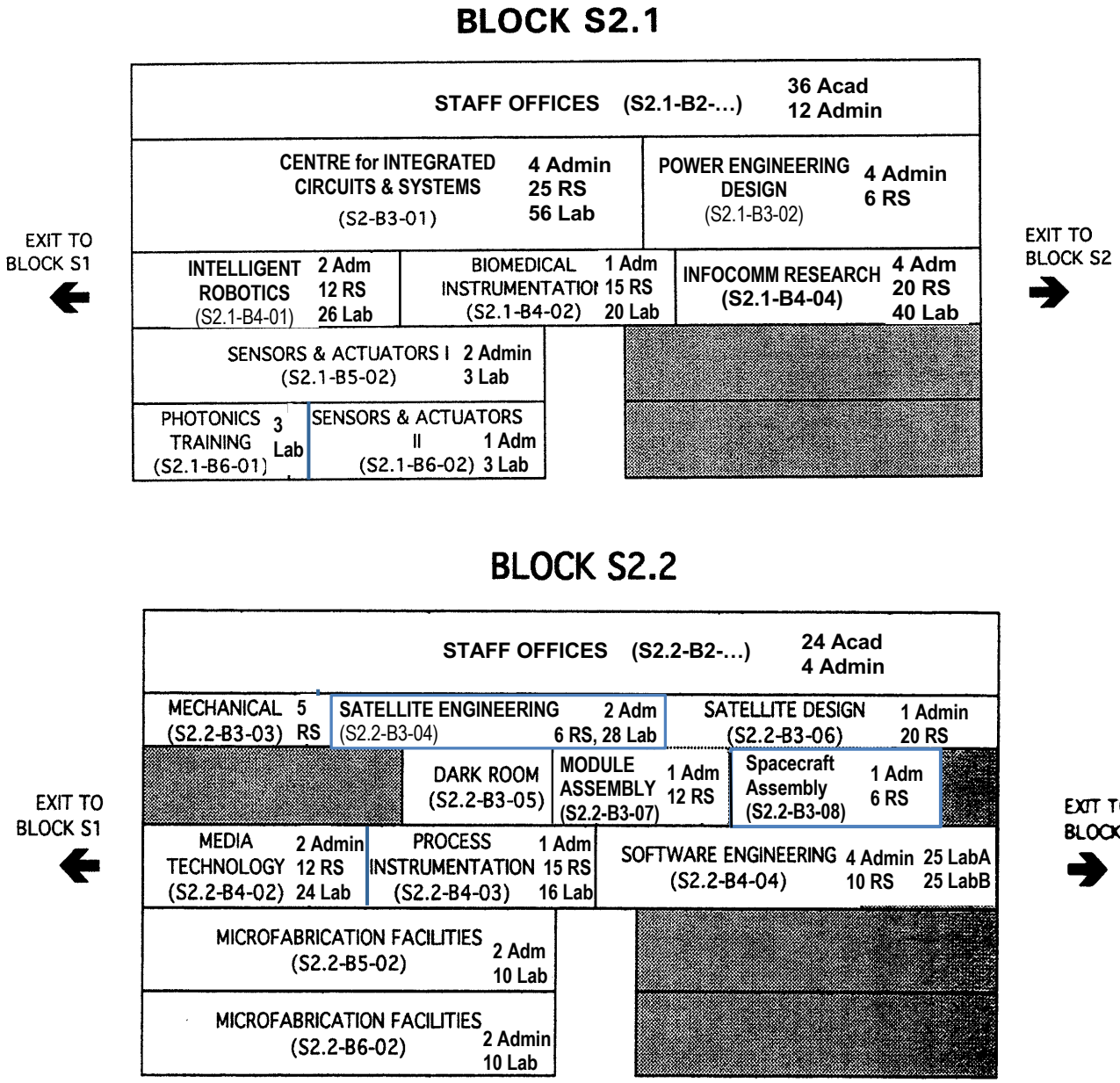

Figure 1. Distribution of PC hosts in labs and offices of S2.1 and S2.2 buildings.

Template for Network Address Assignments using VLSM Subnetting								
Location	Acad		Admin		RS		Lab	
Block S2.1	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits
S2.1-b2	36	x.x.28.01/26	12					
s2.1-b3-01			4		25		56	x.x.30.01/26
s2.1-b3-02			4		6			
s2.1-b4-01			2		12		26	
s2.1-b4-02			1		15		20	
s2.1-b4-04			4		20		40	
s2.1-b5-02			2				3	x.x.30.0010/28
s2.1-b6-01							3	
s2.1-b6-02			1				3	x.x.30.0010/28
Total hosts*	**36+n**		**30**		**78**		**151**	
Block S2.2								
sS.2-b2	24	x.x.28.001/27	4					
s2.2-b3-03					5			
s2.2-b3-04			2		6		28	
s2.2-b3-06			1		20			
s2.2-b3-05								
s2.2-b3-07			1		12			
s2.2-b3-08			1		6			
s2.2-b4-02			2		12		24	
s2.2-b4-03			1		15		16	
s2.2-b4-04			4		10		2x25	
s2.2-b5-02			2				10	
s2.2-b6-02			2				10	
Total hosts*	**24+n**		**20**		**86**		**138**	
* to include additional n addresses for printers, servers, and router interfaces.								

Table 1a. Sample template for network address assignments (contents for illustration only)

Supernetting of Subnets for Network Address Summarization			
Network	Net#/Supernet Bits	Subnets	Description of logical group assignments
x.x.28.0	x.x.28.0/25	x.x.28.01/26 x.x.28.001/27	Supernet for aggregration of Acad groups in S2.1 and S2.2
			<Admin groups>
			<RS groups in S2.1>
			<RS groups in S2.2>
x.x.30.0			Network for all Lab groups in S2.1
x.x.30.32/28			Subnet for combined Lab groups in S2.1-b5-02 & S2.1-b6-02
			<All Lab groups in S2.2>
			<Combined Lab groups in S2.2>

Table 1b. Sample template for network address summarization (contents for illustration only)

User Group	Locations	No of Hosts	No. of Printer/ Server	Sub Network Address	Sub-net Mask	Router Address	First Host Address	Last Host Address	Printer Address	Server Address	No. used / unused
Lab_2.1	CICS	56	1 / 1	207.21.30.0	/26	x.x.x.65	x.x.x 66	x.y.z 121	x.x.x.122	x.x.x.123	59 / 3
	Intelligent Robots	26									
	Biomedical	20									
	Infocommunication Research	40									
	Sensors & Actuators i	3	1/1		2x /29 or /28 ?						2x 6/0 or 12/2 ?
	Sensors & Actuators II	3	1/0								
	Photonics	3									

Table 2. Table template for host address assignments

Server Name	Server IP Address	Subnet Mask	Attached Router Name	Router Interface
HTTP server				
DNS server				
SQL Server				
Acad Server				
Admin Server				
RS Server				
Email Server				
(all ntwk devices)				
Etc…				

Table3. Sample template for server address assignments

Network Name	Router Name	Router Interface	Network #	Subnet mask	IP Address	Access List Number
RS Network	RS S2.1	0/0				x
		1/0				x
		2/0				1 (OUT)
	RS S2.2	0/0				x
		1/0				x
		2/0				x
	RS	0/0				x
		1/0				123 (IN)
		2/0				
		3/0				

Table 4. Sample template for ACL assignments

Network	Router Name	Router Interface	IP Address	Type	Access-List
RS Network	RS S2.2	1/0		In/Out	Extended IP access list 101 **// permit FTP service from RS to Acad** permit tcp x.x.x.x filter y.y.y.y filter eq 21 **// deny FTP service from RS to others** deny tcp .. .. .. any eq 21 **// permit IP access to all** permit ip any any
	RS	0/0		In/Out	Extended IP access list 102 **// permit FTP service from Acad to RS** … **// deny FTP service from others to RS** … **// permit the rest access** …

Table 5. Sample template for access-control list statements